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Research Paper

Optimization and comprehensive evaluation of liquid cooling tank for single-phase immersion cooling data center

Shengchun Liu [a,b], Zhiming Xu [a], Zhiming Wang [a], Xueqiang Li [a,c,*], Haiwang Sun [c], Xinyu Zhang [a], Haoran Zhang [a]

[a] Key Laboratory of Refrigeration Technology of Tianjin, International Centre in Fundamental and Engineering Thermophysics, Tianjin University of Commerce, Tianjin 300134, PR China

[b] Key Laboratory of Efficient Utilization of Low and Medium Grade Energy, Tianjin University, Tianjin 300350, PR China

[c] Tianjin Tier Technology Co., Ltd., Tianjin 300450, PR China

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ABSTRACT

Single-phase immersion cooling has emerged as a highly promising solution for addressing energy challenges and the substantial heat dissipation demands in data centers (DCs). The liquid cooling tank is a curial component in such system where numerous servers are immersed and the coolant are evenly flowed to each server. However, its performance and the comprehensive evaluation for the tank have not received much attention. In this study, by using the validated numerical model, the different tank structures, with or without consideration of baffle and porous panel, were firstly discussed. Then, four key parameters of tank, including inlet flowrate (Parameter A), the ratio of separation chamber height to server height (Parameter B), the ratio of pressure chamber height to server height (Parameter C), and the porosity of the porous panel (Parameter D) were explored in detail. Finally, the tank performance is evaluated and optimized by the orthogonal analysis and the entropy weight method (EWM). Results showed that, flow uniformity within the tank can be enhanced by incorporating a porous panel, and it was advisable to minimize the gaps between servers, despite the potential increase in pressure loss. Parameter A significantly impacted both the maximum temperature and pressure loss. Parameters B and C demonstrated effects on coolant weight and pressure loss, while Parameter D influenced pressure loss. By using orthogonal analysis and the EWM, maximum temperature carried the highest weight (0.362), followed by temperature difference (0.245), coolant weight (0.216), and pressure loss (0.177). The recommended values for Parameters A-D were 4 m^3/h, 1.35 %, 4.05 %, and 11.33 %, respectively, which could be also considered as universality, since the comprehensive evaluation index only changed by 8 % when the server number increased from 10 to 40.

1. Introduction

In recent years, due to the vigorous development of information technology (IT), the number of data centers (DC) has seen explosive growth. This has led to a significant increase in the energy consumption, which consumed 200–250 TW·h of electricity in 2020, equivalent to 1 % of global electricity [1]. By 2025, it is estimated that DCs will account for up to 4.5 % of global power consumption, with 40 % of this energy being used for refrigeration systems [2,3]. Therefore, it is essential to develop more efficient cooling methods.

Currently, the primary cooling method for DCs involves the use of computer room air conditioning (CRAC). However, this method suffers from low energy efficiency due to the limitations of air's thermophysical properties [4]. Additionally, the miniaturization and integration of chips have led to a significant increase in heat flux. Such method has not yet met the higher cooling demands, with a heat dissipation limit of 37 W/cm^2 [5]. To address this issue, single-phase immersion cooling has been considered one of the most promising methods, which can effectively cool chips with high heat flux (up to 100 W/cm^2) [6,7]. According to Wang's work [8], such cooling method can decrease PUE by 20.8 % and 17.6 % compared to air-based cooling method and cold plate cooling method.

As a result, numerous studies have focused on optimizing the single-

* Corresponding author at: Tianjin Key Laboratory of Refrigeration Technology, International Centre in Fundamental and Engineering Thermophysics, Tianjin University of Commerce, 300134, Tianjin, PR China.

E-mail address: xqli@tjcu.edu.cn (X. Li).

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Nomenclature

a	Separation chamber height, mm
b	Pressure chamber height, mm
c	Width of porous panel, mm
c_p	Heat capacity of coolant, J/(kg·°C)
d	Length of porous panel, mm
E	Information entropy
F	Comprehensive evaluation index
g	Gravity acceleration, m/s^2
h	Simulation result
i	Level of parameter
j	Parameter
k	Average value of KPI
L	Server height, mm
m	Coolant weight, kg
n	Number of holes
p	Proportion of the KPI
P	Pressure, Pa
$P_{highest}$	Pressure at the highest point, Pa
P_{in}	Pressure at inlet, Pa
ΔP_{server}	Pressure loss in server, Pa
ΔP	Pressure loss, Pa
q	Inlet flowrate, m^3/h
r	Hole radius, mm
R	Range
s	Serial number of the server
t	Number of servers in tank
T_{max}	Maximum temperature in tank, °C
T_f	Temperature of coolant, °C
T_{server}	CPU maximum temperature in server, °C
T_s	Temperature of solid, °C
ΔT	Temperature difference in tank, °C
u	Velocity in direction x, m/s
v	Velocity in direction y, m/s
V	Volume of tank, m^3
w	Velocity in direction z, m/s
W	Weight, non-dimensional number
Y	Normalization value

Greek symbol

ρ	Density of coolant, kg/m^3
μ	Dynamic viscosity, Pa·s
λ_f	Thermal conductivity of coolant, W/(m·°C)
λ_s	Thermal conductivity of solid, W/(m·°C)

Abbreviations

CDU	Cooling distribution unit
CRAC	Computer room air conditioning
DC	Data center
EWM	Entropy weight method
FVM	Finite volume method
IT	Information technology
KPI	Key performance indicator

phase immersion cooling method. However, the focus of the majority of these studies lies in server-level optimization, encompassing considerations such as server structures, coolants, and the effects of multiple parameters. Numerical simulation emerges as the predominant method. Regarding server structures, optimizing the flow distribution is crucial for enhancing cooling performance. For example, Sarangi et al. [9] recommended using heat sinks with thicker fins and wider channel widths, given the typically high viscosity of coolants. Niazmand et al. [10] observed that pin–fin heat sinks could reduce pressure loss by 50 % compared to micro-channel heat sinks. Muneeshwaran et. al [11] found that compared to traditional heat sink, heat sinks with heat pipe and vapor chamber could decrease CPU temperature by 3.3 °C and 4.8 °C, respectively. Eliand et al. [12] altered the coolant flow path to direct it directly to the CPU, resulting in a 5 °C reduction in CPU temperature. Shrigondekar et al. [13] found the coolant would be bypassed if the server structure was not optimized. In addition to optimizing server structures, the thermophysical properties of the coolant significantly affect cooling performance. Chen et al. [14] discussed the impact of thermophysical property of coolant on cooling performance. Results showed the dynamic viscosity of coolant had the largest influence, followed by thermal conductivity, density, and specific heat capacity. Similar work was also conducted by Hnayno et al. [15]. Nanofluid could also enhance the cooling performance [16–18]. Luo et. al [19] found that the optimal nanofluid concentrations were respectively 0.3 vol% and 3.7 vol% when Reynolds number were under 250 and above 1000. Jeelani et al. [20] and Obalalu et al. [21] found that the nanofluid with different nano-particles had better performance compared nanofluid with single nano-particles. Lastly, when optimizing server structures, it's essential to consider the impact of multiple parameters. Shah et al. [22] observed that CPU temperature reached approximately 62 °C with a 1 L/min inlet flowrate and a 40 °C inlet temperature for 1.5 U servers. Chhetri et al. [23] proposed a regression equation for chip temperature, taking into account chip power, inlet flowrate, and inlet temperature.

In practical operation, dozens of servers are immersed in a liquid cooling tank. The primary purpose of the tank is to ensure uniform flowrates for each server, making comprehensive consideration of tank performance essential; however, the literature is quite limited. For example, Li [24] introduced baffles between the server and the tank to enhance the flow field. The results demonstrated a 1.4 °C reduction in CPU temperature with three baffles compared to none. Saini et al. [25] utilized a distribution manifold to improve the flow field within the tank, resulting in minimal deviation (18.5 %) in flowrates among 10 servers. Ni et al. [26] explored the impact of inlet and outlet locations on tank performance. The findings indicated that positioning the inlet at the tank's bottom and the outlet at the middle could save 11 W of pump power. Apart from thermal performance, it is essential to consider pressure loss and coolant weight when evaluating tank performance.

The entropy weight method (EWM) circumvents the influence of human factors, thereby augmenting the objectivity of comprehensive evaluation outcomes [27,28]. It serves as a tool for ascertaining the weight coefficients of various individual prediction models [29–31]. For example, Shi et al. [32] discussed the weight of maximum temperature, temperature difference, and pack weight for optimizing lithium-ion battery packs, with respective weights of 0.284, 0.266, and 0.45. Based on these results, the overall performance was improved by 60 %.

From the literature review above, it is evident that the performance of liquid cooling tank in single-phase immersion cooling DC has not received much attention, which is used to ensure uniform flow rates for each server. Its performance needs to be optimized and carefully assessed. To bridge this knowledge gap, this study, using numerical simulation, begins by designing and comparing the performance of liquid cooling tanks with and without baffle and porous panel. Following the determination of the tank structure with baffle and porous panel, the impact of four key parameters on tank performance, including inlet flowrate (Parameter A), ratio of separation chamber height to server height (Parameter B), ratio of pressure chamber height to server height (Parameter C), and porosity of porous panel (Parameter D), is studied. Subsequently, tank performance is compressively evaluated by orthogonal design and EWM. Lastly, the versatility and effectiveness of the proposed tank is validated according to change the number of

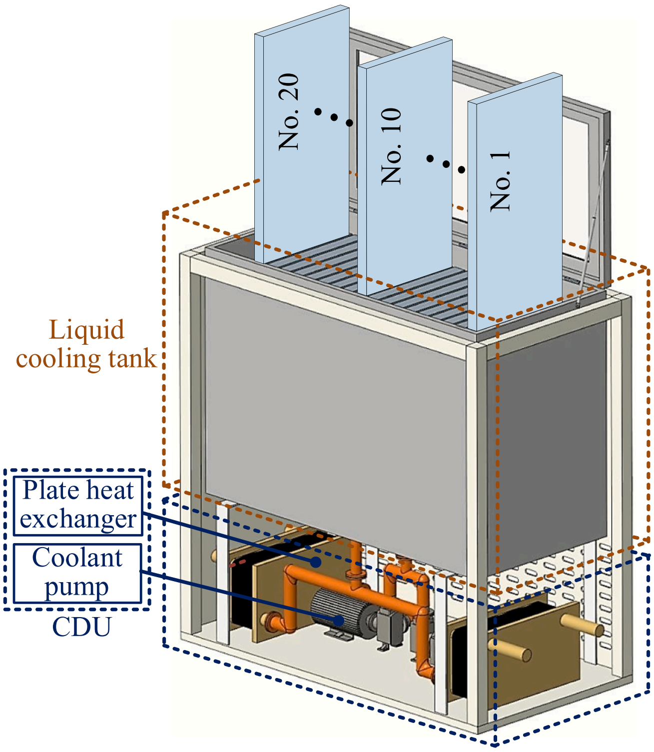


Fig. 1. Schematic of single-phase immersion rack.

Table 1
Thermophysical parameter of coolant.

Temperature (°C)	Density (kg/m^3)	Heat capacity (J/(kg·°C))	Dynamic viscosity (Pa·s)	Thermal conductivity (W/(m·°C))
20	794	1965	0.0117	0.34
40	786	2051	0.0061	0.36
60	775	2154	0.0036	0.38
80	762	2253	0.0024	0.42

servers. The maximum temperature, temperature difference, pressure loss and coolant weight are employed as the key performance indicators (KPI).

2. Model description

For a single-phase immersion cooling rack, it consists of a cooling distribution unit (CDU) and a liquid cooling tank, as shown in Fig. 1. The CDU primarily comprises a coolant pump and a plate heat exchanger. During operation, the coolant pump is employed to propel the coolant into the tank, effectively dissipating the heat. Subsequently, the coolant with high temperature is conveyed to the plate heat exchanger, where it releases the heat to the cooling water. In the liquid cooling tank, 20 servers are arranged. In addition to optimizing the server structure, careful consideration should also be given to the impact of the tank structure in order to enhance the performance of the tank.

The coolant used in this study is "Ice Core 797", which is produced by Tianjin Tier Technology Co. Ltd.. The thermophysical properties of the coolant used in this study can be found in Table 1. This coolant shows the advantages of superior thermophysical properties, exhibit insulative characteristics and being environmentally friendly.

2.1. *Server structures and the equivalent model*

2.1.1. *Description of the server*

The liquid-based cooling server model in this paper is derived from an air-based cooling server (Type: Dell R620, 1U), as illustrated in Fig. 2. This server configuration includes two CPUs, one motherboard, two heat sinks, two power supplies, several hard disks, and memory modules. To enhance coolant flow, three deflectors have been installed on the cover plate, and the heat sinks on the CPUs have been replaced with pin fins [33]. The specific server parameters are listed in Table 2.

2.1.2. *Governing equation, grid, and model validation*

Creo Parametric and FloEFD software are employed to study the server model, which are widely used in refrigeration and heating ventilation air conditioning. The continuity equation, momentum equation, and energy equation of coolant can be found in Eqn. (1)-(3)

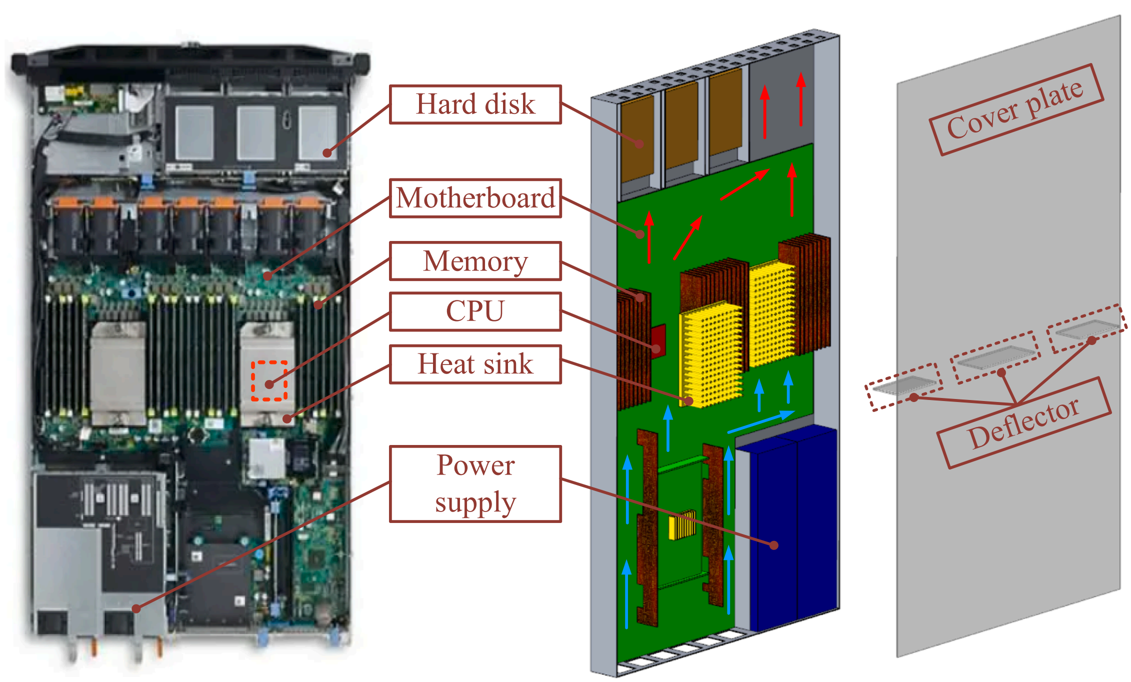


Fig. 2. Server structure.

Table 2
Detail parameters about the server.

Item	Parameter	Size (mm)	Number
Heat sink used in server	Substrate size	120 × 85 × 5	/
	Fin spacing	4.5	/
	Fin height	30	/
	Fin diameter	3	/
Server	CPU (150 W)	40 × 35 × 2	2
	Motherboard	410 × 594 × 2	1
	Memory	140 × 2 × 35	20
	Power supply	230 × 74 × 38	2
	Raid card	24 × 24 × 2	1
	Hard disk	100 × 74 × 6	6

[34,35]. Steady-state analysis is chosen and the governing equations are solved by a discrete numerical technique based on the finite volume method (FVM).

$$\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}=0 \tag{1}$$

$$\begin{cases} \dfrac{\partial(\rho uu)}{\partial x}+\dfrac{\partial(\rho uv)}{\partial y}+\dfrac{\partial(\rho uw)}{\partial z}=-\dfrac{\partial P}{\partial x}+\mu\left(\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}+\dfrac{\partial^2 u}{\partial z^2}\right) \\ \dfrac{\partial(\rho vu)}{\partial x}+\dfrac{\partial(\rho vv)}{\partial y}+\dfrac{\partial(\rho vw)}{\partial z}=-\dfrac{\partial P}{\partial y}+\mu\left(\dfrac{\partial^2 v}{\partial x^2}+\dfrac{\partial^2 v}{\partial y^2}+\dfrac{\partial^2 v}{\partial z^2}\right) \\ \dfrac{\partial(\rho wu)}{\partial x}+\dfrac{\partial(\rho wv)}{\partial y}+\dfrac{\partial(\rho ww)}{\partial z}=-\dfrac{\partial P}{\partial z}+\mu\left(\dfrac{\partial^2 w}{\partial x^2}+\dfrac{\partial^2 w}{\partial y^2}+\dfrac{\partial^2 w}{\partial z^2}\right)-\rho g \end{cases} \tag{2}$$

$$\frac{\partial(\rho uT_f)}{\partial x}+\frac{\partial(\rho vT_f)}{\partial y}+\frac{\partial(\rho wT_f)}{\partial z}=\frac{\partial}{\partial x}\left(\frac{\lambda_f}{c_p}\frac{\partial T_f}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{\lambda_f}{c_p}\frac{\partial T_f}{\partial y}\right)+\frac{\partial}{\partial z}\left(\frac{\lambda_f}{c_p}\frac{\partial T_f}{\partial z}\right) \tag{3}$$

where u, v and w are respectively the velocity in direction x,y and z; ρ, P, μ, T_f, λ_f and c_p are respectively density, pressure, dynamic viscosity, temperature, thermal conductivity and heat capacity of coolant; g is gravity acceleration.

Similarly, the energy equation of solid is computed as follow [36]:

$$\lambda_s\left(\frac{\partial^2 T_s}{\partial x^2}+\frac{\partial^2 T_s}{\partial y^2}+\frac{\partial^2 T_s}{\partial z^2}\right)=0 \tag{4}$$

where λ_s and T_s are respectively thermal conductivity and temperature of solid.

To improve the accuracy, the grids of heat sink and CPU are refined, as shown in Fig. 3(a). Through the grid independence test, it can be found that the variation of CPU maximum temperature was changed slightly when the grid number exceeded 122.2×10^4, as illustrated in Fig. 3(b). Therefore, such grid number was employed.

To validate the server model, the data from Chhetri's work was employed [23]. The server validation model can be found in Fig. 4(a). During the operation, the inlet temperature and heat load respectively maintained at 40 °C and 65 W, respectively. The inlet flowrate increased from 0.5 L/min to 2.5 L/min. The validation results are shown in Fig. 4 (b). It can be found the deviation was less than 2 °C and therefore, it can be considered as credible.

2.1.3. Equivalent model of the server

Based on the studied server (Fig. 2), the variations of the CPU maximum temperature and the pressure loss in server along with the variation of inlet flowrate can be obtained by using the validated model, as shown in Eqns. (5) and (6). Therefore, by using these two equations, the sever can be simplified as the equivalent model to simulate the performance of liquid cooling tank. In the equivalent model, the electronic components, such as the CPU, heat sink, motherboard, etc., are simplified [37,38], as shown in Fig. 5.

$$T_{server}=73.7+0.606/q \tag{5}$$

$$\Delta P_{server}=7.21+193q-462.9q^2+398.8q^3 \tag{6}$$

where q is the inlet flowrate.

2.2. Designed tank structures

2.2.1. Description of the designed tank structures

The main function of the liquid cooling tank is mainly to distribute the coolant to servers evenly. Therefore, the different structures are designed, as shown in Fig. 6(a)-(c). The main differences for these three structures lie in the arrangement of baffle and porous panel. The function of the baffle ensures all the coolant flowing through the server, and the porous panel to distribute coolant evenly to servers. For structure 1, both the baffle and porous panel are considered, as shown in Fig. 6(a). For structure 2, only the porous panel is included, as shown in Fig. 6(b). And for structure 3, only the baffle is used, as illustrated in Fig. 6(c). Although there may have other tank structures, such as the different inlet and outlet locations, the above three structures are employed due to they would affect the flow distribution largely. More detailed parameters about the tank can be also found in Table 3.

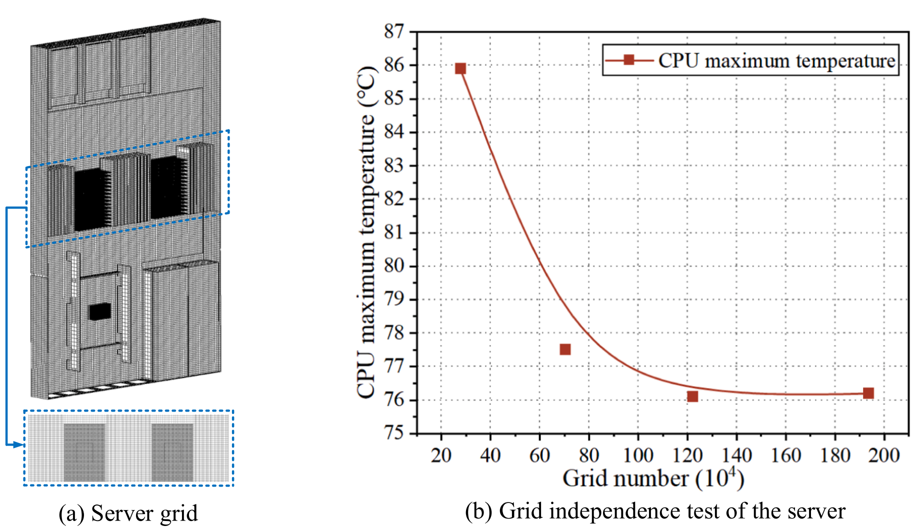
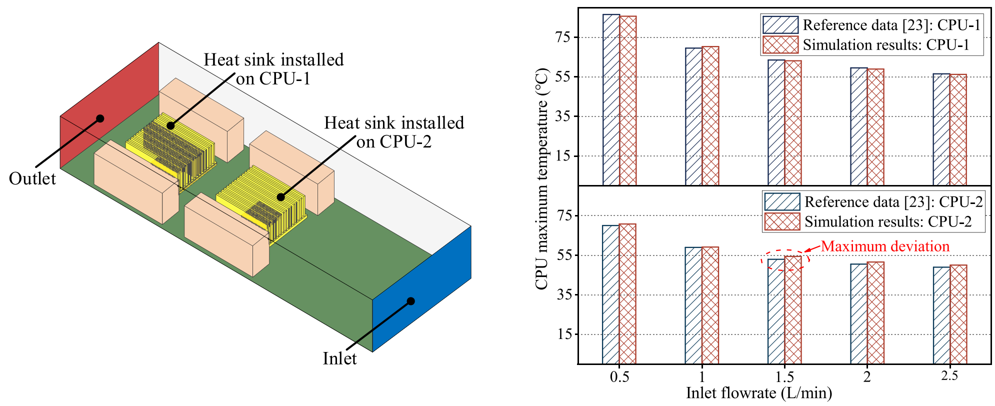


Fig. 3. Grid details and gird independence test of the server.



(a) Validation model (b) Model validation

Fig. 4. Structure of validation model and model validation.

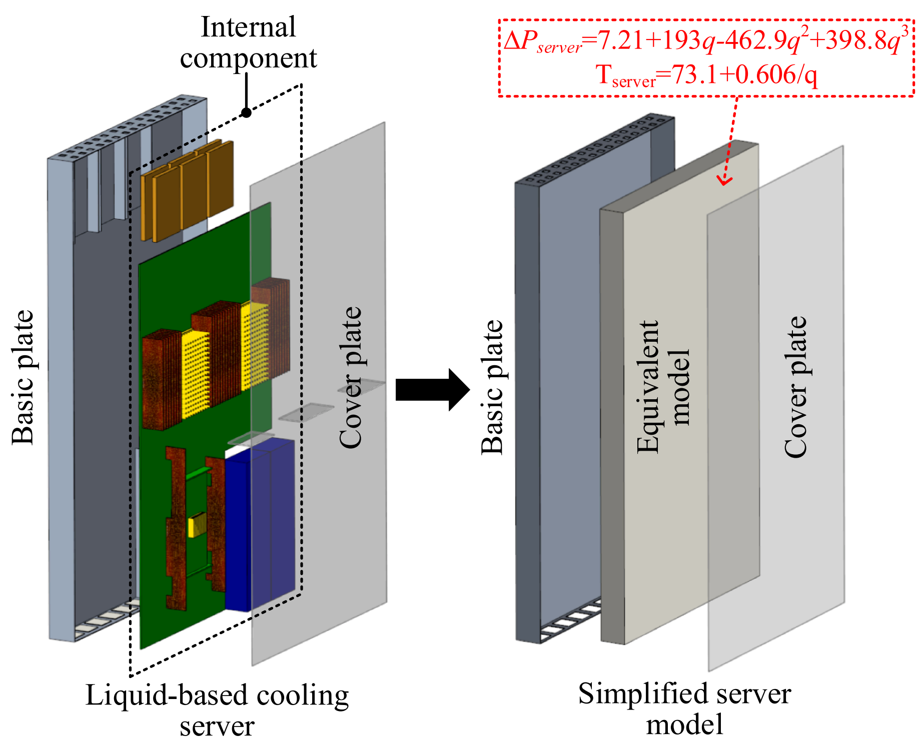
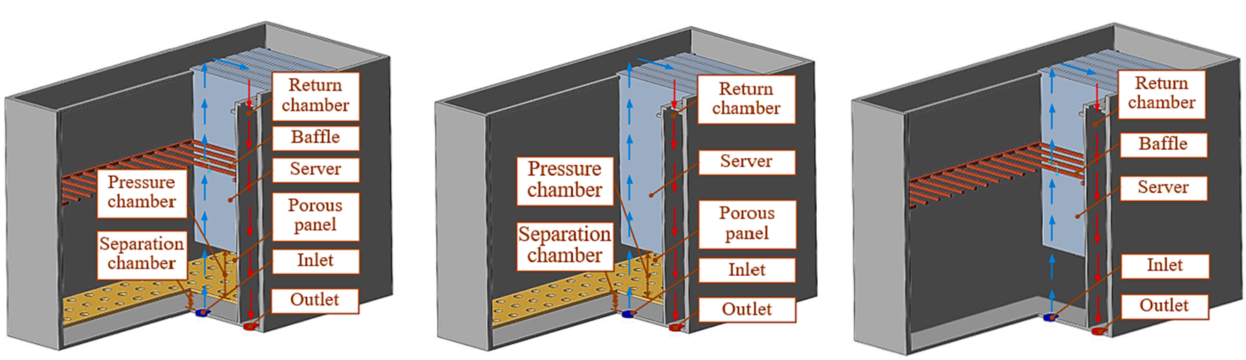


Fig. 5. Equivalent model of the server.



(a) Designed structure 1 (Tank structure with consideration of baffle and porous panel) (b) Designed structure 2 (Tank structure with consideration of porous panel) (c) Designed Structure 3 (Tank structure with consideration of baffle)

Fig. 6. Schematic of different designed tank structures.

According to the mechanism of heat transfer and flow, four key parameters are considered, including the inlet flowrate (A), ratio of separation chamber height to server height (B), ratio of pressure chamber height to server height (C), and porosity of porous panel (D). The inlet flowrate (A) would largely affect the server temperature and pressure loss. Parameters B-D could affect the tank volume and flow distribution for each server, which also needs to be considered. The schematic for each parameter and the detailed parameter can be found in Fig. 7 and Table 4.

2.2.2. Governing equation and boundary conditions

When simulating the performance of liquid cooling tank, the equivalent model of the server is employed. The continuity equation is the same with Eqn. (1). A source term was added into the momentum equation to simulate the pressure loss in the server, as shown in Eqn. (7). The heat transfer process during the simulation is simplified by Eqn. (5). The detailed boundary conditions are listed in Table 5.

$$\begin{cases} \frac{\partial(\rho uu)}{\partial x}+\frac{\partial(\rho uv)}{\partial y}+\frac{\partial(\rho uw)}{\partial z}=-\frac{\partial P}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right)+\Delta P_{server} \\ \frac{\partial(\rho vu)}{\partial x}+\frac{\partial(\rho vv)}{\partial y}+\frac{\partial(\rho vw)}{\partial z}=-\frac{\partial P}{\partial y}+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right)+\Delta P_{server} \\ \frac{\partial(\rho wu)}{\partial x}+\frac{\partial(\rho wv)}{\partial y}+\frac{\partial(\rho ww)}{\partial z}=-\frac{\partial P}{\partial z}+\mu\left(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\right)-\rho g+\Delta P_{server} \end{cases} \tag{7}$$

2.2.3. Grid independence test

The grid details of tank structure are illustrated in Fig. 8(a) and the porous panel is refined to simulate the flow field more accurately. The grid independence test can be found in Fig. 8(b). It is observed that as the grid number increased from 238 × 10^4 to 371 × 10^4, there was a significant reduction in both temperature difference and pressure loss. However, beyond a grid number of 541.6 × 10^4, the variations in these parameters became negligible, stabilizing at approximately 0.1 °C and 9.9 Pa, respectively. Consequently, a grid number of 541.6 × 10^4 was selected.

2.2.4. Model validation of tank model

To validate the accuracy and reliability of tank model considering the equivalent model, the experiment was also conducted. Maximum temperatures were monitored by using AVIDA64 software, which can directly measure CPU diode temperatures. The experimental setup is depicted in Fig. 9. During the experiment, the inlet temperature and CPU heat load were maintained at 40 °C and 150 W, respectively. The inlet flowrate ranged from 0.1 m^3/h to 0.5 m^3/h. Detailed parameters of experiment condition are shown in Table 6.

The results of model validation can be found in Fig. 10 in which the data at steady state is employed. The simulation results closely aligned with the experimental data, showing a consistent trend. The maximum temperature decreased with increasing inlet flowrate. Furthermore, the temperature deviation between the experimental and simulation results were within 0.8 °C. Thus, the proposed tank model considering the equivalent model exhibits the capability to predict tank performance in single-phase immersion cooling scenarios.

Table 3
Detail parameters about the liquid cooling tank.

Item	Parameter	Size (mm)	Number
Liquid cooling tank	Server	420 × 44 × 741	20
	Tank	1096 × 496×(776 + a + b)	1
	Baffle	1090 × 490 × 5	1
	Porous panel	1090 × 490 × 5	1
	Inlet radius	20	1
	Outlet radius	20	1

2.3. Key performance indicators (KPIs)

To accurately assess the tank performance, four KPIs, namely, maximum temperature, temperature difference, pressure loss, and coolant weight, were considered. The maximum temperature is the highest CPU temperature among the 20 servers in the tank, which should remain below 85 °C, as shown in the following:

$$T_{\max} = Max(T_{server,s}|s = 1,2,3,\cdots,t) \tag{8}$$

where s is the serial number of the server and t is the number of servers in tank.

Temperature difference is used to measure flow uniformity, which can be calculated as follows:

$$\Delta T = Max(T_{server,s}|s = 1,2,3\cdots,t) - Min(T_{server,s}|s = 1,2,3\cdots,t) \tag{9}$$

Pressure loss is the difference between inlet pressure and the highest point pressure, as shown in the following:

$$\Delta P = P_{in} - P_{highest} \tag{10}$$

where P_{in} and $P_{highest}$ are pressure at the inlet and the highest point, respectively.

The coolant weight serves as a pivotal indicator of tank lightweight design. The reduction in coolant weight not only enables the building to accommodate a greater number of racks within its structural capacity, but also leads to significant cost savings, which is shown in the following:

$$m = \rho V \tag{11}$$

where V is tank volume, respectively.

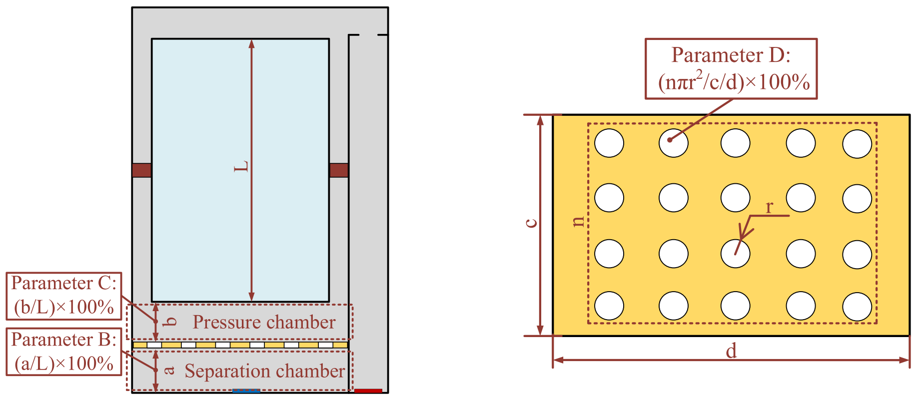
3. Results and discussion

3.1. Performance comparison of designed tank structures

To fairly compare the performance of different tank structures, parameter A-D are assumed to be constant, which are respectively 4 m^3/h, 9.45 %, 4.05 %, and 0.29 %, respectively. Through Fig. 11, it is clear that structure 1 demonstrated the lowest maximum temperature and temperature difference, primarily because the employment of baffle and porous panel ensured even coolant distribution within the server. When comparing structures 1 and 2, it is evident that the incorporation of baffle redirected the coolant flow through the servers rather than through the gaps between servers. This redirection resulted in a decrease of 0.9 °C in maximum temperature and 0.3 °C in temperature difference, despite potentially causing a slight increase in pressure loss. The result indicated the server gaps should be avoided in the practical application. When comparing structure 1 and structure 3, the influence of the porous panel became more apparent. The addition of the porous panel resulted in a reduction of maximum temperature and temperature difference by 1.2 °C and 3.1 °C, respectively, but it also increased the pressure loss by 428.6 Pa. Moreover, the variation in coolant weight among these three tank structures was expected to be negligible, even with the inclusion of baffles and porous panels, which may occupy some volume. Given that the primary concern in DCs is ensuring the safe operation of servers, both baffles and porous panels should be considered, even though they may lead to an increase in pressure loss. Therefore, structure 1 was employed to do the following analysis.

3.2. Effect of parameters on performance of designed tank structure 1

Fig. 12 illustrates the influence of the inlet flowrate (Parameter A) on the tank's performance. Evidently, an increase in the flowrate facilitated more efficient heat removal, leading to a significant reduction in the maximum temperature. However, this also resulted in an increase in



(a) Schematic of A-A view (b) Schematic of porous panel

Fig. 7. Schematic of detailed structure.

pressure loss. Therefore, the flowrate needed to be carefully considered in practical application, as shown in Fig. 12(a). For instance, when the flowrate was raised from 1 m^3/h to 10 m^3/h, the maximum temperature decreased from 86 °C to 76.3 °C, representing a decrease of 9.7 °C. However, this came at the cost of increased pressure loss and temperature difference, which can rise from 6629.6 Pa to 11030.1 Pa and 0.23 °C to 1.84 °C, respectively. Furthermore, the coolant weight remained constant due to the unchanging tank volume, as shown in Fig. 12(b). Therefore, it is imperative to conduct a comprehensive assessment when considering the impact of flowrate.

Fig. 13 illustrates the impact of the ratio of separation chamber height to server height (Parameter B) on the tank's performance. It can be observed that this ratio had a minor influence on the maximum temperature and temperature difference, indicating that the overall flow pattern remained unchanged. However, it had an effect on pressure loss and coolant weight. Since an increase in Parameter B resulted in an enlargement of the tank volume, the coolant weight increased linearly. Within the studied range (1.35 % to 9.45 %), the coolant weight varied by 11.6 %. Additionally, there existed an optimal ratio that minimized pressure loss, which, in this case, was found to be 3.5 %. This can be attributed to the fact that at smaller ratios, high-speed coolant flow directly impacted the porous panel, leading to increased pressure loss. Conversely, at larger ratios, gravity must be overcome, resulting in higher pressure loss.

Fig. 14 demonstrates the influence of the ratio of pressure chamber height to server height (Parameter C) on the tank's performance. As seen in Fig. 14(a), this parameter also exhibited a marginal effect on both maximum temperature and temperature difference. However, it notably influenced pressure loss and coolant weight. As Parameter C increased, the tank volume expanded, leading to a linear increase in both pressure loss and coolant weight, indicating this ratio should be as small as possible, as shown in Fig. 14(b). For instance, when this ratio increased from 1.35 % to 9.45 %, there was a corresponding increase of 6.6 % in pressure loss and 10.8 % in coolant weight.

Fig. 15 illustrates the impact of the porosity of the porous panel (Parameter D) on the tank's performance. It is evident that this ratio has a significant effect on pressure loss. For the studied range (0.29 % to 20 %), the pressure loss decreased with the increase of porosity, which was due to the increasing drag forces at lower porosity. When the porosity increased from 0.29 % to 3.05 %, the pressure loss decreases by 416.8 Pa. However, when it increased from 3.05 % to 20 %, the reduction of pressure loss is not obvious, which is only 9.7 Pa. In addition, it's worth noting that porosity has a minor influence on maximum temperature, temperature difference, and coolant weight.

3.3. Comprehensive evaluation for designed tank structure 1

In order to comprehensively assess the performance of the liquid cooling tank, all four aforementioned parameters must be thoroughly considered. In this study, orthogonal analysis is used to assess these four parameters on the performance, the EWM is used to comprehensive evaluate the tank performance.

For orthogonal analysis, a standard $L16(4^4)$ orthogonal design scheme is developed. The detailed information is provided in Table 7.

Range analysis is the most commonly method to analyze the orthogonal results, in which R_j can be used to determine the priority order of the degree of influence of factors on KPIs [40], which can be obtained through the following equation:

$$R_j = \max(k_{i,j}) - \min(k_{i,j}) \tag{12}$$

where k is the average value of the KPI, i is the level of parameter, j is the parameter.

Besides the range analysis, the four KPIs also needs to be considered to comprehensively assess the liquid cooling tank performance. EWM is employed in this study, which is a common weighting method to mea-

Table 4
Range of the four key parameters.

Item	Parameter	Equation	Unit	Range
A	Inlet flowrate	/	m^3/h	1–10
B	Ratio of separation chamber height to sever height	(a/L) × 100 %	%	1.35–9.45
C	Ratio of pressure chamber height to sever height	(b/L) × 100 %	%	1.35–9.45
D	Porosity of porous panel	$(n\pi r^2/c/d) \times 100$ %	%	0.29–16.85

Table 5
Boundary conditions for simulation.

Item	Content	Parameter
Inlet	Volume flowrate (m^3/h)	1–10
Outlet	Environment pressure (Pa)	101,325
Ambient parameter	Temperature (°C)	40
	Gravity (z direction) (m/s^2)	−9.81
Server	Equivalent model	/
Tank, baffle and porous panel	Real wall	/

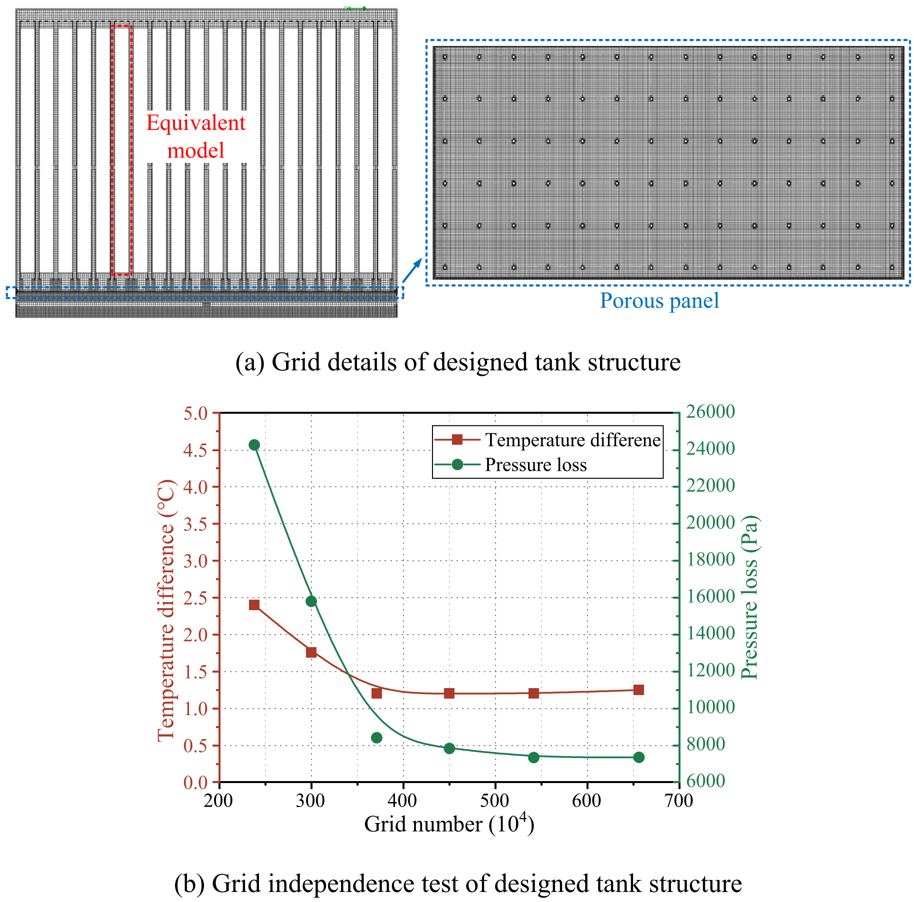


Fig. 8. Grid details and independence test of designed tank structure.

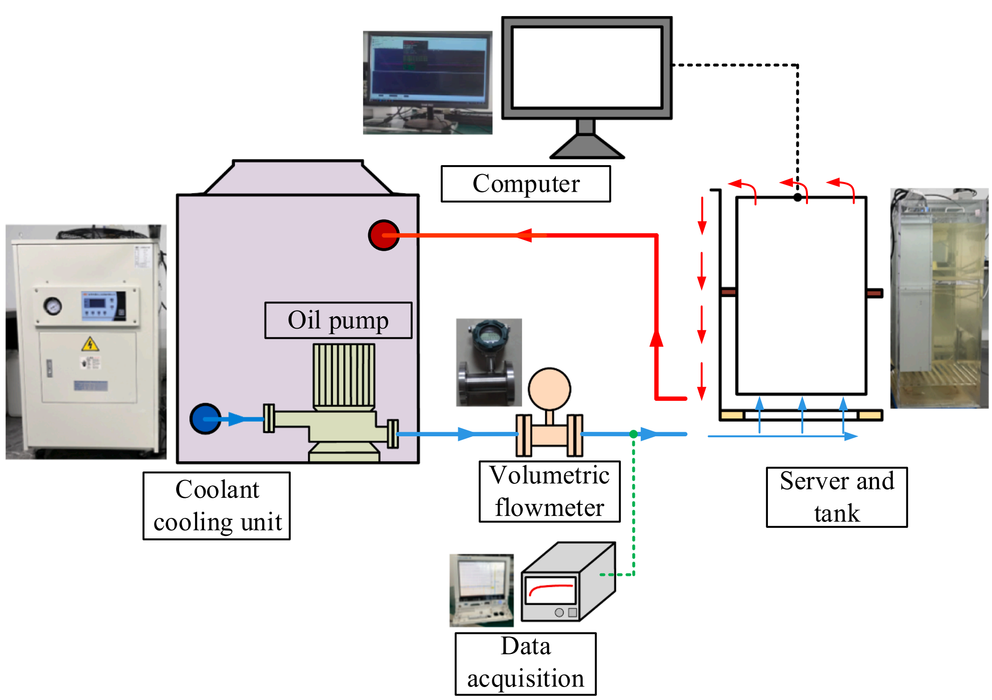


Fig. 9. Experiment system.

Table 6
Detailed information of experiment.

Items	Content	Parameter
Experiment device	Coolant cooling unit	Temperature control range: −30–100 °C Accuracy: ± 0.1 °C
	Oil pump	Rate flowrate: 1.1 m^3/h Variable frequency motor regulation
	Volumetric flowmeter	Flowrate range: 0.05–0.75 m^3/h Accuracy: 1 %
	Liquid cooling tank	Size: 525 × 365 × 1100 mm
	Data acquisition	Type: Yokogawa-GP10
	Server	Reference on Table 2
	Thermocouples	Type: TT-K-30 Temperature range: −60–300 °C
Test environment	Inlet flowrate	0.1–0.5 m^3/h
	CPU heat load	150 W
	Inlet temperature	40 °C

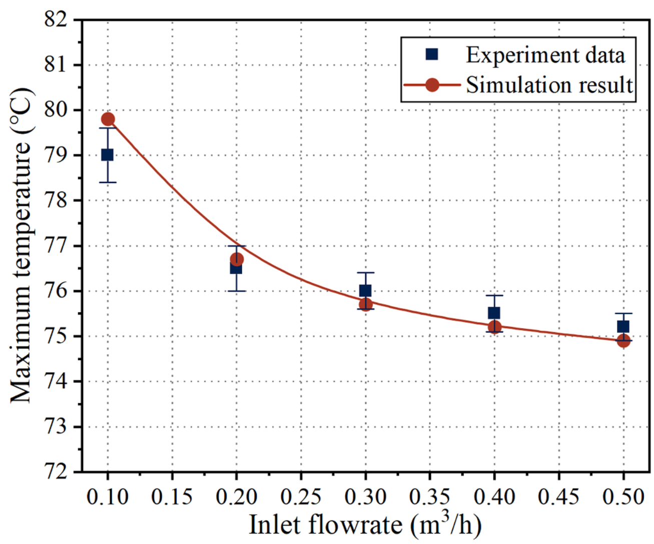


Fig. 10. Model validation about liquid cooling tank.

sure the dispersion in decision making. The greater the degree of dispersion, higher weight is obtained. The comprehensive evaluation index (F) is employed to assess the tank's performance, which can be calculated as follows [40]:

$$F_s = \sum_{KPI=1}^{4} W_{KPI} Y_{s,KPI} \tag{13}$$

where s represents case s in orthogonal experiment; W and Y are weight of each KPI and normalization value. They can be obtained in the following [41–43]:

$$Y_{s,KPI} = \frac{\max(h_{KPI}) - h_{s,KPI}}{\max(h_{KPI}) - \min(h_{KPI})} \tag{14}$$

$$\begin{cases} W_{KPI} = \dfrac{1 - E_{KPI}}{\sum_{KPI=1}^{4}(1 - E_{KPI})} \\ E_{KPI} = \dfrac{\sum_{s=1}^{16} p_{s,KPI} \cdot \ln p_{s,KPI}}{\ln 16} \\ p_{s,KPI} = \dfrac{Y_{s,KPI}}{\sum_{s=1}^{n} Y_{s,KPI}} \end{cases} \tag{15}$$

where h, E, and p represent the simulation result in orthogonal experiment, information entropy, and proportion of a specific value in all results, respectively.

Based on the orthogonal experiment, 16 cases were simulated. The simulation results, depicted in Fig. 16, reveal the impacts of the four parameters on four KPIs. Concerning the maximum temperature, Parameter A (inlet flowrate) exhibited the most significant influence, rendering the impact of the other parameters negligible in practical applications. For temperature difference, all four parameters had an influence, with Parameter A having the largest impact, followed by Parameters D, C, and B. In the case of pressure loss, Parameter B had the least influence, followed by Parameters D, C, and A. Notably, only Parameters B and C needed to be considered, as changes in Parameters A and D didn't affect the tank volume or coolant weight.

Based on the simulation results, the weights for each KPI were calculated using EWM, as illustrated in Fig. 17(a). Among these four KPIs, maximum temperature exhibited the highest weight (0.362),

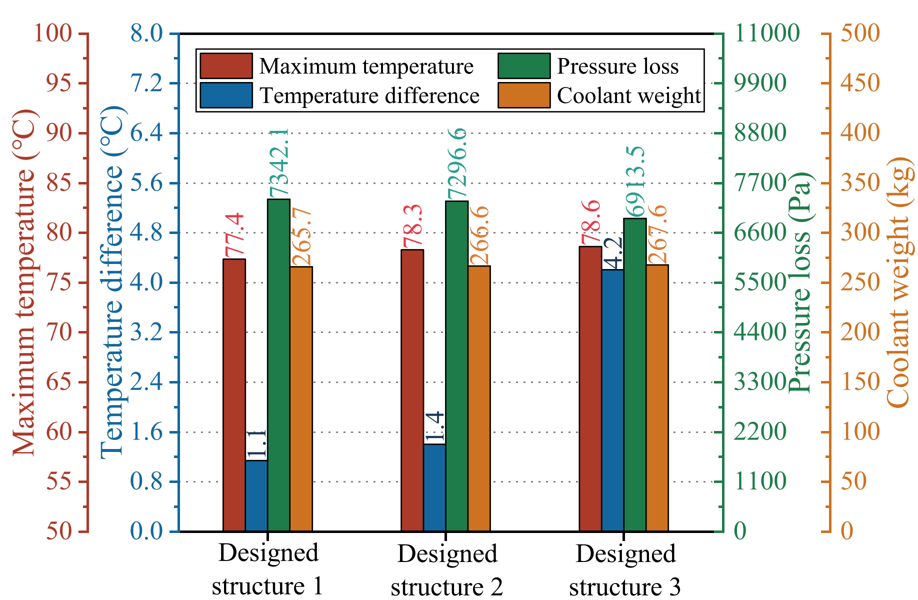
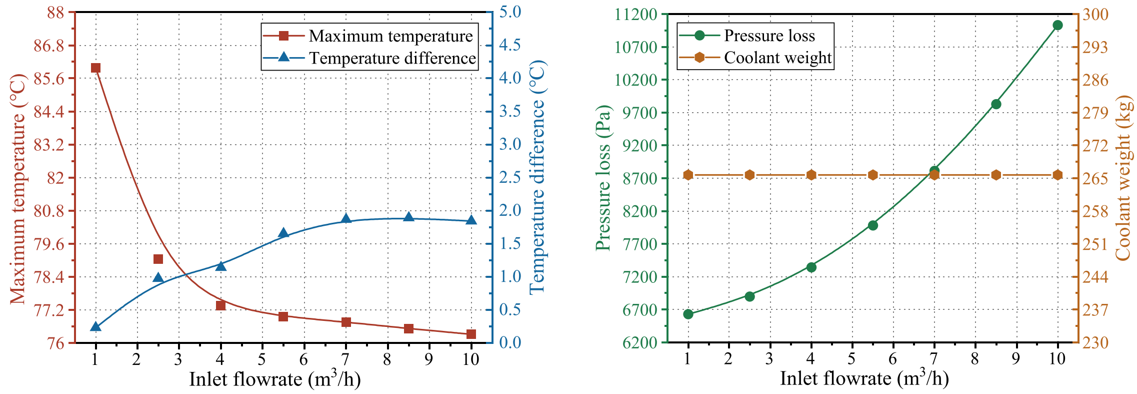


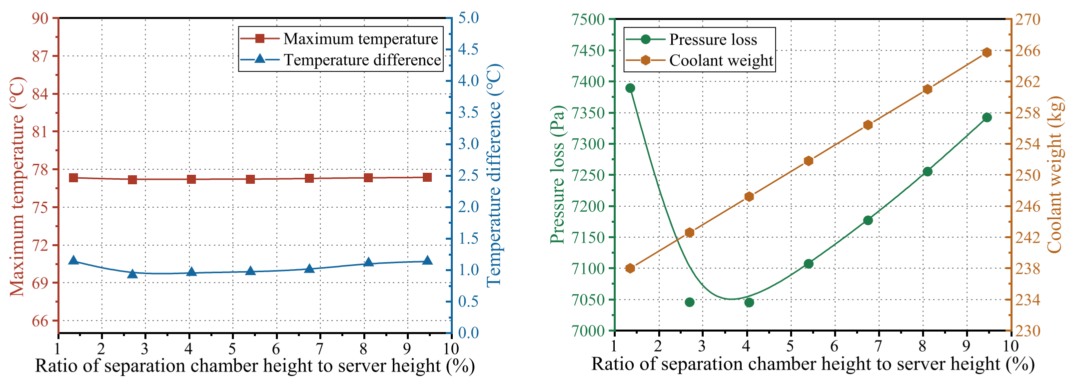
Fig. 11. Performance comparison of different tank structures.



(a) Impact of Parameter A on maximum temperature and temperature difference

(b) Impact of Parameter A on pressure loss and coolant weight

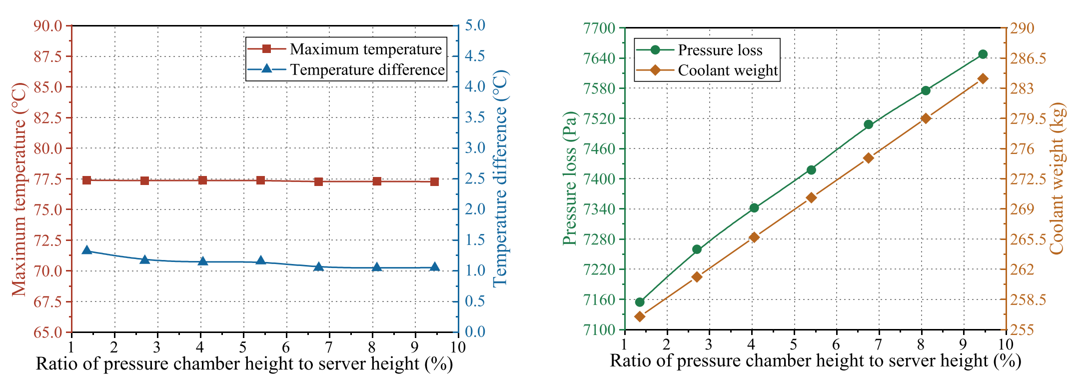
Fig. 12. Impact of inlet flowrate on tank performance.



(a) Impact of Parameter B on maximum temperature and temperature difference

(b) Impact of Parameter B on pressure loss and coolant weight

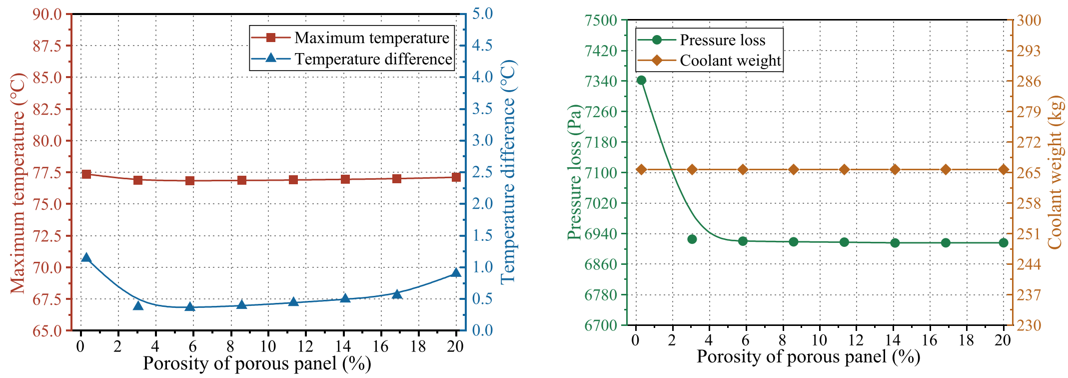
Fig. 13. Impact of ratio of separation chamber height to server height on tank performance.



(a) Impact of Parameter C on maximum temperature and temperature difference

(b) Impact of Parameter C on pressure loss and coolant weight

Fig. 14. Impact of ratio of pressure chamber height to server height on tank performance.



(a) Impact of D on maximum temperature and temperature difference

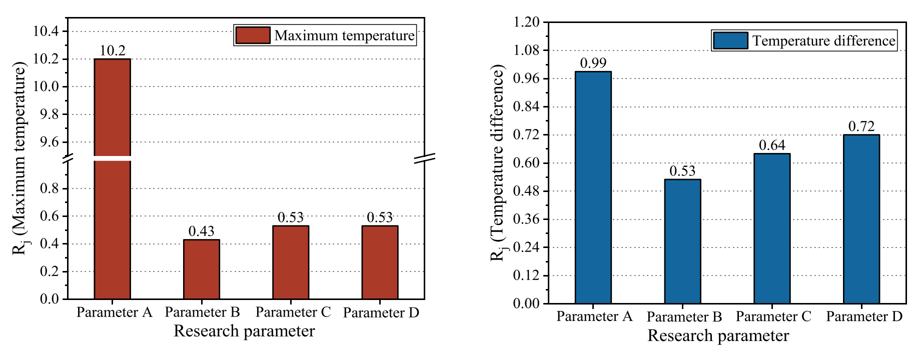
(b) Impact of D on pressure loss and coolant weight

Fig. 15. Impact of porosity of porous panel on tank's performance.

Table 7
Orthogonal design schematic [39].

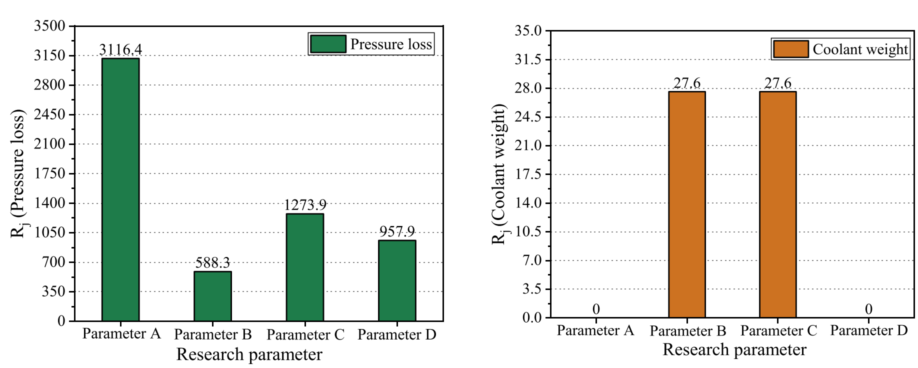
Case	A (m^3/h)	B (%)	C (%)	D (%)	Case	A (m^3/h)	B (%)	C (%)	D (%)
1	1	1.35	1.35	0.29	9	7	1.35	6.75	16.85
2	1	4.05	4.05	5.81	10	7	4.05	9.45	11.33
3	1	6.75	6.75	11.33	11	7	6.75	1.35	5.81
4	1	9.45	9.45	16.85	12	7	9.45	4.05	0.29
5	4	1.35	4.05	11.33	13	10	1.35	9.45	5.81
6	4	4.05	1.35	16.85	14	10	4.05	6.75	0.29
7	4	6.75	9.45	0.29	15	10	6.75	4.05	16.85
8	4	9.45	6.75	5.81	16	10	9.45	1.35	11.33

Annotation: A: inlet flowrate, B: Ratio of separation chamber height to sever height, C: Ratio of pressure chamber height to sever height, D: Porosity of porous panel.



(a) Range analysis on maximum temperature

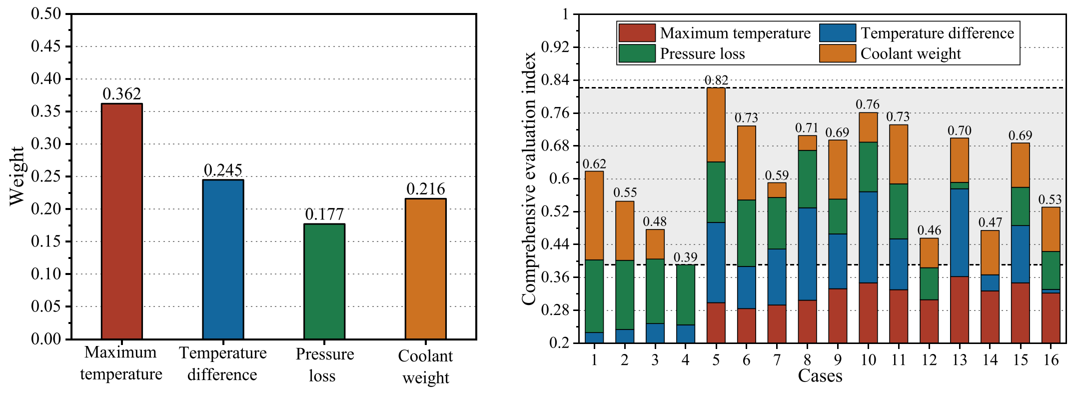
(b) Range analysis on temperature difference



(c) Range analysis on pressure loss

(d) Range analysis on coolant weight

Fig. 16. Range analysis on tank performance.



(a) Weight of different KPIs (b) Comprehensive evaluation index

Fig. 17. Comprehensive evaluation index of orthogonal cases.

emphasizing its primary consideration in practical applications. This was followed by temperature difference (0.245), coolant weight (0.216), and pressure loss (0.177). These weightings offered valuable guidance in the design of the single-phase immersion cooling system. Additionally, the results of the orthogonal experiment were also analyzed using EWM, as depicted in Fig. 17(b). It is evident that Case 5 had the highest comprehensive evaluation index, scoring 0.82, representing a 52.4 % increase compared to Case 4. Consequently, Case 5 was recommended, the specific values for Parameters A-D were 4 m^3/h, 1.35 %,4.05 %, and 11.33 %, respectively.

3.4. Universality validation

To assess the universality of the suggested case, simulations were conducted for varying numbers of servers. In accordance with the number of servers, both tank volume and inlet flowrate were proportionally varied, while other parameters remained constant. The results are presented in Fig. 18. It is clear that, the overall score decreased with the increase of server number, which was primarily attributed to a decrease in the maximum temperature and temperature difference scores, indicating the flow patterns within the tank became less uniform with a larger number of servers, as shown in Fig. 19. Moreover, the comprehensive evaluation index for pressure loss and coolant weight

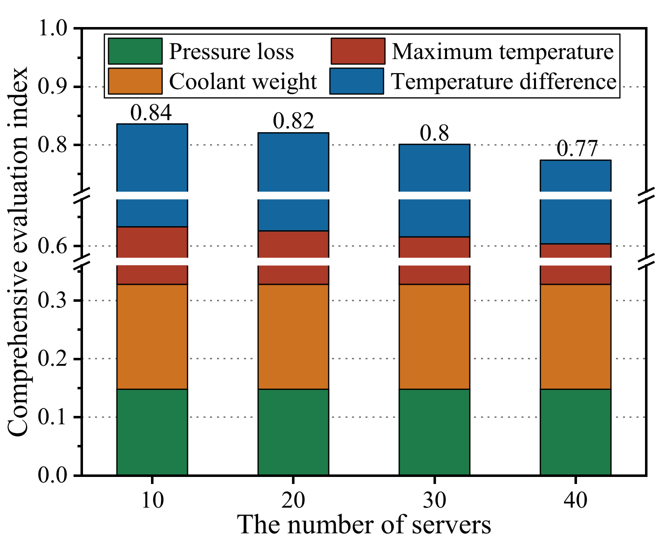


Fig. 18. Comprehensive evaluation index of different server numbers.

remained unchanged, which was mainly due to proportional changed in the tank volume and the coolant weight as the number of servers varied. It is noteworthy that the comprehensive evaluation index consistently remained within an ideal range, decreasing from 0.84 to 0.77 with the server number increased from 10 to 40; corresponding, the maximum temperature and temperature difference increased from 76.8 °C to 77.5 °C and 0.3 °C to 1.1 °C. This suggests that the recommended case can be considered universal.

4. Conclusions and future work

This study focuses on optimizing and comprehensive evaluating the performance of liquid cooling tank in single-phase immersion cooling data center by using a validated numerical model. Firstly, different tank structures are discussed, followed by the four key parameters, including inlet flowrate (A), the ratio of separation chamber height to server height (B), the ratio of pressure chamber height to server height (C), and the porosity of the porous panel (D). Then, the tank performance is evaluated by using orthogonal analysis and the entropy weight method (EWM). Based on the findings, the following conclusions can be drawn:

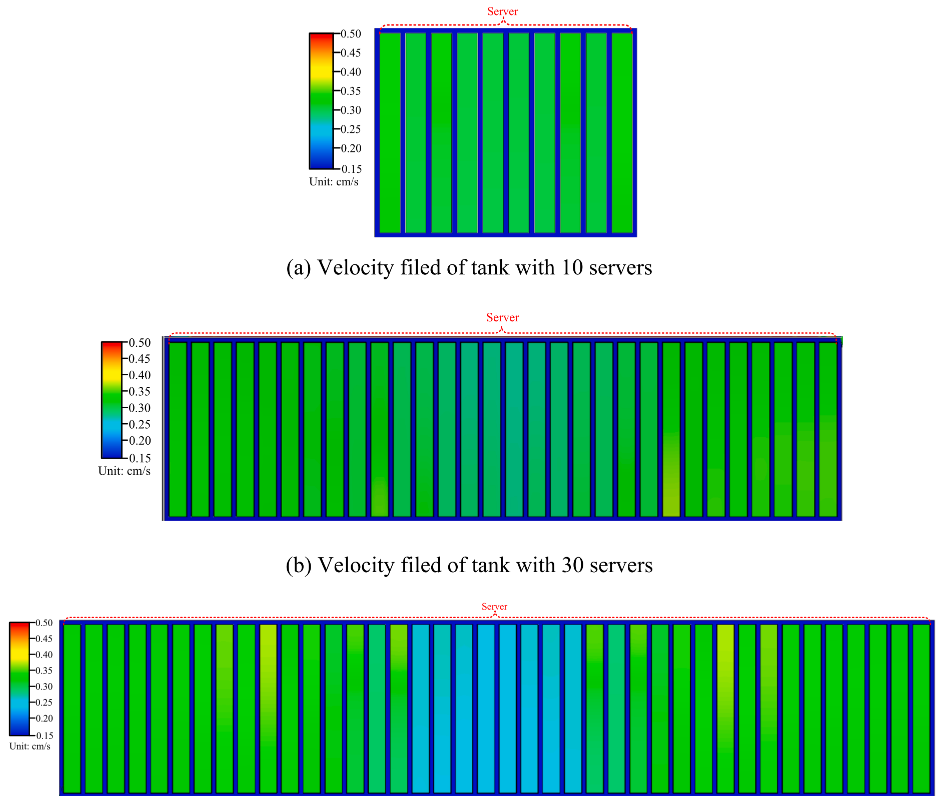
(1) During practical application, the flow uniformity in the tank can be improved by using porous panel. In addition, the gap between servers should be avoided.

(2) Parameter A significantly impacts both the maximum temperature and pressure loss. Parameters B and C demonstrate effects on coolant weight and pressure loss, while Parameter D influences pressure loss.

(3) By using orthogonal analysis and the EWM, maximum temperature carried the highest weight (0.362), followed by temperature difference (0.245), coolant weight (0.216), and pressure loss (0.177), which could provide a guidance when designing the liquid cooling tank.

(4) Recommended values for Parameters A-D in liquid cooling tank were suggested to be 4 m^3/h, 1.35 %, 4.05 %, and 11.33 %, respectively. This structure could be also considered as universality, since the comprehensive evaluation index only changed by 8 % when the server number increased from 10 to 40; correspondingly, the maximum temperature and temperature difference increased by 0.7 °C and 0.8 °C, respectively.

The recommended tank structure and parameter are provided in this study. However, during the practical application, the heat load of server may be changed and therefore, the corresponding adjustment of energy consumption devices is also needed, which includes the coolant pump, cooling water pump, and the fan in the single-phase immersion cooling system. Besides, the environment should be also considered to comprehensively assess the variation of energy efficiency of such system. This work will be conducted in the following work based on the



(a) Velocity filed of tank with 10 servers

(b) Velocity filed of tank with 30 servers

(c) Velocity filed of tank with 40 servers

Fig. 19. Velocity filed of tanks with different sever numbers.

recommended tank structure.

CRediT authorship contribution statement

Shengchun Liu: Writing – review & editing, Methodology, Formal analysis, Data curation. **Zhiming Xu:** Writing – review & editing, Investigation, Data curation. **Zhiming Wang:** Writing – review & editing, Investigation, Data curation. **Xueqiang Li:** Investigation, Data curation. **Haiwang Sun:** Project administration. **Xinyu Zhang:** Formal analysis. **Haoran Zhang:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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